

Notes: Hau et al. (JF, 2024)

FinTech Credit and Entrepreneurial Growth

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1 Big picture

- **Question.** Does access to FinTech credit improve the growth of small firms, and if so, which firms benefit most and through what operational channels?
- **Setting.** The paper studies Taobao vendors in China that can obtain automated credit lines from Ant Group. Ant uses transaction data from Alibaba's platform plus other financial information to compute a credit score. Around a cutoff score of 480, the probability of credit approval jumps discontinuously, which creates a fuzzy regression discontinuity design.
- **Core challenge.** Firms that receive credit are not randomly chosen. Better firms may both qualify for credit and grow faster. The paper therefore needs a source of quasi-exogenous variation in credit access and also has to show that firms cannot precisely manipulate the running variable around the cutoff.
- **Main contribution.** The paper combines a platform-wide universe of vendors with a discontinuity in the credit decision rule. This lets the authors estimate the causal effect of credit *access* on growth, customer ratings, and operating choices for a very large sample of small firms.
- **Main baseline result.** Access to a FinTech credit line raises sales growth by about 44% and transaction growth by about 31% over the two-month window from $t - 1$ to $t + 1$. Product, service, and consignment ratings each rise by about 0.054, roughly one quarter of a sample standard deviation.
- **Main heterogeneity result.** The gains are larger for younger firms, firms in industries with more dispersed growth, firms in less durable-goods sectors, and firms less likely to own real estate. This pattern is the paper's strongest evidence that FinTech credit has an information advantage over traditional bank credit.
- **Main mechanism result.** Credit access is followed by more advertising, greater product variety, and a higher visitor-to-purchaser conversion rate. These operating changes provide a concrete link from financing to growth and customer satisfaction.
- **Preliminary market-structure result.** In cities where state-bank density is high and SOE output share is high, more firms enter Taobao and more FinTech credit is supplied and used. The paper interprets this as evidence that traditional bank structure crowds out private entrepreneurs and that FinTech partly fills that gap.
- **Economic takeaway.** The paper is not just about a new lending technology. It is about how data-rich credit provision can relax entrepreneurial financing frictions in a setting where traditional banks favor the state sector and collateral-rich borrowers.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- Alibaba's Taobao platform generates high-frequency transaction data for millions of small online vendors.

- Ant Group uses these data to provide automated small-business credit, avoiding the need for a physical branch network and a traditional relationship-loan officer model.
- The paper studies a part of China’s economy where small firms often face severe credit frictions, especially because state-owned banks channel a large share of credit toward SOEs.
- A key empirical advantage is that the data include the full platform population rather than only loan applicants, so the paper can compare credit-eligible and credit-ineligible firms inside the same ecosystem.

Interpretation. The setting is ideal for asking whether a new credit technology can expand entrepreneurial growth beyond the reach of traditional bank lending.

2.2 Ant Group’s credit approval process and the 480 cutoff

- Ant computes a continuous credit score that runs roughly from 380 to 680.
- For most months in the main sample, a vendor is generally approved for a credit line when the score exceeds 480.
- Credit eligibility is reassessed monthly in an automated process. Once a vendor is approved, the online contract takes about three minutes to complete.
- The credit line behaves like a revolving credit product. Maturity is usually 12 months, and at least 1/12 of the drawn balance must be repaid each month.

Two data features make the design *fuzzy* rather than sharp:

- **No-show cases.** Some firms above 480 are still denied credit because of additional hard exclusions such as prior default, bad platform conduct, dubious suppliers, or conflict-of-interest rules.
- **Cross-over cases.** The researchers observe end-of-month scores, while Ant generally bases approval on beginning-of-month information. Occasional within-month score changes create some approved firms with an observed lagged score below 480.

Interpretation. The cutoff still creates powerful quasi-experimental variation, but the probability of treatment rises from about 0.33 below the threshold to roughly 0.63 just above it rather than jumping from zero to one.

2.3 Samples

The paper uses two main samples.

City-level sample.

- Built from Taobao vendors with location information, which is available for about one-third of the full platform population.
- Used to study whether platform entry and aggregate FinTech credit are higher in places where traditional local credit supply is more distorted.
- Entry is reconstructed from 2005 to 2015 using vendor establishment dates.

Firm-month FRDD sample.

- Main outcome sample covers November 2014 to June 2015; December 2014 is excluded because the 480 rule was briefly suspended.
- Treated firms must have access to credit at the end of month t and still have access at the end of month $t + 1$.
- Control firms must have no observable credit access at either the end of month t or the end of month $t + 1$.
- Restricting attention to firms with scores in $[460, 500]$ leaves 1,196,887 observations for 547,491 firms.

Interpretation. The treatment definition intentionally requires persistence over two months so that the estimated effect is not diluted by status-switching right after the cutoff event.

2.4 City-level variables

The preliminary cross-sectional analysis uses three main outcomes:

- $\ln(\# \text{ New TB Firms})$: log number of new Taobao entrants from a city,
- $\ln(\text{Credit Line})$: log total monthly credit line offered by Ant Group in the city,
- $\ln(\text{Credit Use})$: log total monthly drawn balance used by vendors in the city.

The main city-level regressors are:

- **Loan/GDP**: overall local banking development,
- **State Bank Density**: number of state-bank branches per 10,000 people,
- **SOE Output Share**: local importance of the state sector,
- plus local income, population, and digital-development controls.

Interpretation. This part of the paper is not the causal core. It provides background evidence that FinTech credit is especially relevant where state-dominated banking leaves entrepreneurs underserved.

2.5 Firm-level outcome variables

The main firm-month outcomes are:

$$\text{Sales Growth}_{i,t+1} = \log(\text{Sales}_{i,t+1}) - \log(\text{Sales}_{i,t-1}), \quad (1)$$

$$\text{Transaction Growth}_{i,t+1} = \log(\text{Transactions}_{i,t+1}) - \log(\text{Transactions}_{i,t-1}). \quad (2)$$

Customer-satisfaction outcomes come from Taobao’s Detailed Seller Ratings system:

- **Product Rating**: quality and match with product description,
- **Service Rating**: responsiveness and helpfulness of the vendor,

- **Consignment Rating:** delivery speed and shipping quality.

Each rating is winsorized and standardized to the $[0, 1]$ interval for comparability.

Interpretation. The paper is careful not to focus only on sales. It also studies customer capital — a form of intangible value — through ratings that are directly tied to the buyer experience.

2.6 Mechanism variables

To open the black box between financing and growth, the paper studies three additional outcomes:

- **Advertisement:** $\ln(1 + \text{monthly advertising expenditure})$,
- **Product Variety:** $\ln(1 + \text{number of product types})$,
- **Customer Conversion Rate:** completed purchasers divided by page visitors (scaled by 100).

Interpretation. These variables capture decisions that can plausibly be changed quickly after credit access, which is why they are useful for explaining rapid improvements in sales and ratings.

2.7 Key heterogeneity measures

The paper studies three main dimensions of FinTech’s comparative advantage.

Information advantage.

- **Age Rank:** younger firms are more opaque to traditional lenders.
- **High Dispersion:** industries with higher cross-firm growth dispersion are harder to evaluate using coarse traditional information.

Collateral substitution.

- **High Durability:** durable-goods sectors proxy for stronger pledgeable asset value.
- **Property Ownership:** indicator for high estimated probability that the vendor owns real estate.

Other channels.

- **Distribution Cost:** inverse of predicted credit-line size, meant to capture how costly traditional small-ticket lending would be.
- **Legal Quality:** city-level confidence in the legal system, used in the appendix to study enforcement.

2.8 Descriptive facts

Table I gives the basic scale of the data.

- In the city sample, the average city has about 3,887 new Taobao firms and mean credit line of 9,932 (in ¥10,000 units).

- In the FRDD sample, mean sales growth is -0.44 , mean transaction growth is -0.48 , and the average credit score is 484.7.
- The mean approval rate is 62%.
- The mean credit line is RMB 15,907, with a median of RMB 10,000.
- Average monthly sales for these firms are about RMB 32,337, so the line size is economically meaningful relative to operating scale.

Interpretation. These are genuinely small firms. The results should therefore be read as evidence on working-capital-type credit for small entrepreneurial vendors rather than on large corporate finance.

3 Models and empirical specifications (all equations + interpretation)

3.1 Cross-sectional entry and credit regression

For the city-level analysis, the paper's logic can be summarized by

$$Y_c = \alpha + \beta_1(StateBankDensity_c \times SOEShare_c) + \beta_2 StateBankDensity_c + \beta_3 SOEShare_c + \Gamma X_c + \varepsilon_c, \quad (3)$$

where Y_c is either log new Taobao entry, log credit line, or log credit use.

Interpretation. The coefficient on $StateBankDensity \times SOEShare$ asks whether state-dominated banking is associated with more online entrepreneurial entry and more FinTech credit. A positive coefficient means that FinTech entry is strongest where traditional local credit is least entrepreneur-friendly.

3.2 Baseline first stage: fuzzy approval probability

The paper estimates the probability of getting credit with

$$CreditApproval_{i,t} = \alpha + \rho IV_{i,t} + \sum_{k=1}^K \gamma_k S_{i,t}^k + \phi_j + \phi_t + \varepsilon_{i,t}, \quad (4)$$

where:

- $IV_{i,t} = 1$ if $CreditScore_{i,t} \geq 480$,
- $S_{i,t} = CreditScore_{i,t} - 480$,
- ϕ_j are firm-type fixed effects,
- ϕ_t are month fixed effects.

The baseline uses a local linear model ($K = 1$) on the score range $[460, 500]$.

Interpretation. The coefficient ρ is the first-stage jump in approval probability at the threshold. In Table III it is 0.2872, so crossing 480 raises the chance of treatment by about 28.7 percentage points.

3.3 Baseline second stage: local average treatment effect

The second stage is

$$Dep_{i,t+1} = \beta + \tau \widehat{CreditApproval}_{i,t} + \sum_{k=1}^K \lambda_k S_{i,t}^k + \phi_j + \phi_t + \varepsilon_{i,t}, \quad (5)$$

where Dep is one of the five main firm outcomes.

Interpretation. The coefficient τ is the local average treatment effect of credit access for firms around the threshold. Because the design is fuzzy, τ is the ratio of the reduced-form jump in outcomes to the first-stage jump in approval probability.

3.4 Graphical discontinuity logic

Before going to 2SLS, the paper plots binned averages around the cutoff.

- Firms just above 480 are about 14 percentage points higher in sales growth than firms just below 480.
- Over the same score window, the approval probability rises by about 33 percentage points.

A back-of-the-envelope Wald ratio gives

$$0.14/0.33 \approx 0.42, \quad (6)$$

which is close to the formal baseline effect on sales growth.

Interpretation. The figure already contains the central message of the paper: the discontinuity is economically large, not just statistically significant.

3.5 Heterogeneity specification

To test whether FinTech matters more where information frictions are worse, the paper estimates

$$CreditApproval_{i,t} = \alpha + \rho IV_{i,t} + \delta_1 InfoAdv_{i,t} + \delta_2 IV_{i,t} \times InfoAdv_{i,t} + \sum_{k=1}^K \gamma_k S_{i,t}^k + \phi_j + \phi_t + \varepsilon_{i,t}, \quad (7)$$

$$CreditApproval_{i,t} \times InfoAdv_{i,t} = \alpha + \rho IV_{i,t} + \delta_1 InfoAdv_{i,t} + \delta_2 IV_{i,t} \times InfoAdv_{i,t} + \sum_{k=1}^K \gamma_k S_{i,t}^k + \phi_j + \phi_t + \varepsilon_{i,t}, \quad (8)$$

$$Dep_{i,t+1} = \beta + \tau \widehat{CreditApproval}_{i,t} + \gamma_1 InfoAdv_{i,t} + \gamma_2 \widehat{CreditApproval}_{i,t} \times InfoAdv_{i,t} + \sum_{k=1}^K \lambda_k S_{i,t}^k + \phi_j + \phi_t + \varepsilon_{i,t}. \quad (9)$$

The same structure is then reused with collateral proxies and other channel variables in place of $InfoAdv$.

Interpretation. The interaction coefficient is the key object. It asks whether the causal effect of FinTech credit is larger for firms whose credit quality is harder to evaluate or whose collateral is weaker.

3.6 Why the paper studies credit access rather than only drawdowns

The treatment is *credit approval*, not loan utilization. The authors emphasize that a credit line may alter firm behavior even without immediate drawdown because it acts as liquidity insurance and relaxes precautionary motives.

Interpretation. This matters conceptually. The paper estimates the value of *available financing capacity*, not just the effect of a realized loan balance.

3.7 Estimation details

- The preferred bandwidth is $[460, 500]$, with robustness to $[465, 495]$ and $[470, 490]$.
- Standard errors are clustered at the firm-type level.
- The paper also checks alternative functional forms: different left and right slopes and second-order polynomials.
- For longer-horizon outcomes, it redefines treatment and control to require more persistent credit-status differences over time.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline effect

Identification comes from the jump in approval probability at score 480. Firms very near the cutoff are assumed to be similar in all smooth determinants of growth, while differing discretely in treatment probability.

Interpretation. This is a local experiment. The paper does not claim to identify the effect of credit for all firms in China; it identifies the effect for firms near Ant's approval threshold.

4.2 Why the design is fuzzy rather than sharp

The score threshold does not map perfectly into approval because of no-show and cross-over cases. That is why the paper uses 2SLS rather than a simple before/after or above/below comparison.

Interpretation. The instrument is the rule-based discontinuity, not the realized approval indicator itself.

4.3 Why manipulation of the running variable is unlikely

The paper argues that vendors cannot easily game the score because:

- the exact scoring rule is unknown to them,
- vendors do not observe their own score,

- the score density is smooth around 480,
- and a formal manipulation test of Cattaneo, Jansson, and Ma cannot reject continuity of the density ($p = 0.8796$).

Interpretation. This is one of the strongest credibility checks in the paper. If firms could manipulate the forcing variable, the RDD would be much less convincing.

4.4 Why the sample avoids credit-demand selection

Unlike many loan datasets, the paper observes all Taobao vendors, not just firms that apply for credit.

Interpretation. This means the main comparison is not “borrowers who applied and got approved” versus “borrowers who applied and got denied.” It is “platform firms with and without credit access near the threshold.” That is much cleaner because the sample is not conditioned on unobserved demand for a loan.

4.5 Why timing helps causal interpretation

The data are monthly. The observed running variable is the lagged end-of-month score, while outcomes are measured after approval. In addition, the paper requires treatment persistence from month t through month $t + 1$.

Interpretation. The timing is set up to reduce attenuation from temporary status reversals and to make it harder to interpret the results as contemporaneous reverse causality.

4.6 Fixed effects and what they remove

At the firm-month level, the model includes firm-type and month fixed effects. At the city level, the cross-sectional regressions include rich observable controls for development, bank penetration, population, and digital infrastructure.

Interpretation. The city-level exercise is descriptive. The causal core is the FRDD, where the fixed effects ensure that the cutoff is not merely proxying for monthly sectoral or macro conditions.

4.7 Robustness and placebo exercises

The main robustness checks are:

- three-month treatment windows,
- change rather than level of ratings,
- different left and right slopes,
- second-order score polynomials,
- narrower bandwidths,
- and a placebo test using 1,000 random fake cutoffs.

In the placebo exercise, the average first-stage jump at fake cutoffs is -0.0005 , with a median of -0.001 , far from the true first stage of about 0.29 .

Interpretation. The placebo is especially useful because it attacks the exact source of identifying power. The paper shows that there is nothing special about arbitrary nearby cutoffs — only the real institutional threshold matters.

4.8 What the paper cannot fully identify

The paper is transparent about two limitations.

- It cannot fully separate the effect of mere line availability from the effect of actual borrowing and spending.
- It cannot tell whether higher sales are partly demand diversion from rival vendors or a true increase in aggregate demand, so aggregate welfare and macro growth are harder to assess than firm-level outcomes.

Interpretation. The causal effect on firm-level growth is convincing. The leap from that to aggregate efficiency or welfare is more speculative.

5 Tables and figures

The paper's empirical story is very structured. It begins with the approval discontinuity, then establishes large causal effects on growth and customer ratings, then shows that the effect is stronger exactly where a FinTech information advantage should matter most, and finally links the growth effect to concrete operating adjustments.

5.1 Figure 1: the credit-score cutoff is economically meaningful

Read:

- Panel A shows the score distribution and the discrete jump in approval at 480.
- Panel B shows that approval probability rises by roughly 30 percentage points around the cutoff.
- The score density looks smooth around 480, which visually supports the no-manipulation claim.

Economic message. This figure is the core design. Without it, there is no quasi-experiment. With it, the paper has a credible local source of exogenous credit access.

5.2 Table I and Table II: the firms are small, and the background banking story matters

Read:

- Table I shows that the FRDD sample firms are small: average credit line is RMB 15,907 and average monthly sales are RMB 32,337.

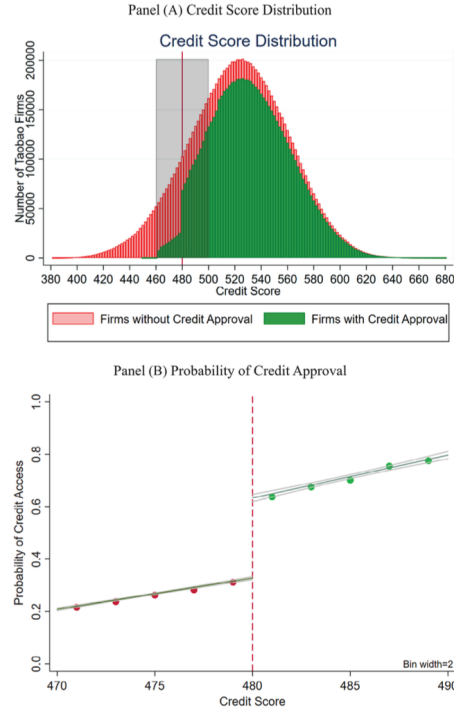


Figure 1. Credit score distribution and credit approval. Panel A plots the distribution of monthly credit scores for 2 million Taobao firms from November 2014 to June 2015. The green bars mark the Taobao firms with credit approval, and the (upper) red bars represent those without credit approval. The gray-shaded region marks the sampling interval for the FRDD with the discontinuity at the credit score of 480. Panel B is the discontinuity plot for the probability of credit approval against credit scores. The vertical axis is the probability of credit access. The horizontal axis is the credit score in the local range of [470, 490]. Each dot on the figure represents the average probability that a credit line is granted to a firm located in the credit score range with a bandwidth of two. The probability is estimated by dividing the total number of firms with credit access by the total number of eligible firms in the same bin. A linear line is fit to the scattered dots on each side of the cutoff score (i.e., 480), surrounded by a 95% confidence interval in light gray lines. (Color figure can be viewed at wileyonlinelibrary.com)

**Table I
Summary Statistics**

This table presents summary statistics for the variables used in our two main regression samples. Panels A and B report summary statistics for all variables in the city-level sample and FRDD sample, respectively.

	<i>N</i>	<i>P10</i>	<i>P50</i>	<i>Mean</i>	<i>P90</i>	<i>SD</i>
Panel A. City-Level Sample						
<i>No. New TB Firms</i>	267	103	542	3,887.44	8361	12,774.03
<i>Credit Line</i> (in ¥10,000)	273	62.15	838.35	9,932.22	18,210.44	38,790.36
<i>Credit Use</i> (in ¥10,000)	273	12.13	144.64	1,920.20	3,758.10	7,291.76
<i>Loan/GDP</i> (%)	267	41.89	59.93	71.28	115.65	36.85
<i>State Bank Density</i>	267	0.36	0.72	0.88	1.50	0.64
<i>SOE Output Share</i>	267	0.01	0.08	0.15	0.41	0.17
<i>PCGDP</i> (in ¥10,000)	267	0.55	1.13	1.72	3.43	2.19
<i>Population</i> (in million)	267	1.41	3.54	4.11	7.41	2.42
<i>Digital Development Index</i>	267	92.29	105.74	124.73	212.3	47.73
Panel B. Local FRDD Sample						
<i>Sales Growth</i>	1,196,887	-2.48	-0.36	-0.44	1.49	2.02
<i>Transaction Growth</i>	1,196,887	-2.48	-0.47	-0.48	1.43	1.61
<i>Product Rating</i>	1,196,887	0.27	0.60	0.57	0.85	0.22
<i>Service Rating</i>	1,196,887	0.28	0.62	0.59	0.87	0.23
<i>Consignment Rating</i>	1,196,887	0.27	0.61	0.58	0.86	0.23
<i>Credit Score</i>	1,196,887	467.72	486.90	484.70	497.80	11.02
<i>Credit Approval</i>	1,196,887	0	1	0.62	1	0.48
<i>Credit Line</i>	1,196,887	0	10,000	15,907.16	18,000	57,647.37
<i>Age Rank</i>	1,196,848	0.04	0.12	0.16	0.34	0.13
<i>High Dispersion</i>	1,196,205	0	1	0.75	1	0.43
<i>High Durability</i>	1,196,887	0	0	0.12	1	0.33
<i>Property Ownership</i>	1,196,887	0	0	0.20	1	0.40
<i>Ln (1 + Ad Expense)</i>	1,196,887	0	0	2.94	8.46	3.71
<i>Ln (1 + Product Types)</i>	1,196,887	2.20	3.78	3.79	5.50	1.35
<i>Customer Conversion Rate</i>	1,196,887	1.08	5.18	7.44	16.07	7.56

**Table II
Macro Factors Affecting China's E-Commerce Entrepreneurship and FinTech Credit**

This table presents cross-sectional regression results on the macro factors influencing the aggregate entry rate and FinTech credit access of Taobao firms at the city level. *# New TB Firms* denotes the total number of newly registered Taobao firms in a city from 2005 to 2015, where only firms with city information available are counted. *Credit Line* measures the total amount of eligible credit lines approved by Ant Group for all Taobao firms (with location information) in a city as of an average month in 2015. *Credit Use* is the total outstanding balance of credit drawn by all Taobao firms (with location information) in a city as of an average month in 2015. Independent variables are measured in the earliest year available during 2005 to 2015 for the entry regression or with the latest information for the 2015 credit access regression. *PCGDP* is the GDP per capita of a city in 10,000 RMB. *Digital Development Index* is a province-level variable developed by Guo et al. (2020). *Loan/GDP* is the ratio of commercial loans to GDP in a city in percentage points. *State Bank Density* is the ratio of the number of state bank branches to the total population (in 10,000) in a city. *SOE Output Share* is the share of output contributed by SOEs in a city, where SOE is defined following the registration type in the Annual Survey of Industrial Firms in China. We report *t*-statistics based on robust standard errors in parenthesis. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Dependent Var.	<i>Ln(# New TB Firms)</i> (1)	<i>Ln(Credit Line)</i> (2)	<i>Ln(Credit Use)</i> (3)
<i>State Bank Density</i> × <i>SOE Output Share</i>	1.6282*** (2.96)	1.7612*** (2.59)	1.4501* (1.84)
<i>State Bank Density</i>	0.2125 (1.30)	-0.0090 (-0.03)	0.0339 (0.11)
<i>SOE Output Share</i>	-0.2368 (-0.80)	-0.7390 (-1.39)	-0.4487 (-0.36)
<i>Loan/GDP</i> (%)	0.0046*** (2.73)	0.0018 (0.79)	0.0016 (0.76)
<i>PCGDP</i> (in millions)	0.2417*** (2.97)	0.1403*** (3.28)	0.1346*** (2.83)
<i>Population</i> (in millions)	0.3679*** (13.63)	0.3303*** (3.89)	0.3390*** (3.93)
<i>Digital Development Index</i>	0.0460*** (9.92)	0.0509*** (9.08)	0.0522*** (8.86)
<i>N</i>	267	273	273

- Mean credit score is 484.7, right near the identification region, and 62% of firm-months in the local sample have credit approval.
- Table II shows that the interaction of state-bank density and SOE output share predicts more Taobao entry and more FinTech credit. The coefficient is 1.628 for log new Taobao firms and 1.761 for log credit line.

Economic message. Table I tells you what kind of firms are being studied. Table II explains why this setting is important: FinTech seems to expand where the traditional local banking structure is least supportive of private entrepreneurs.

5.3 Figure 2 and Table III: the baseline causal effect is large

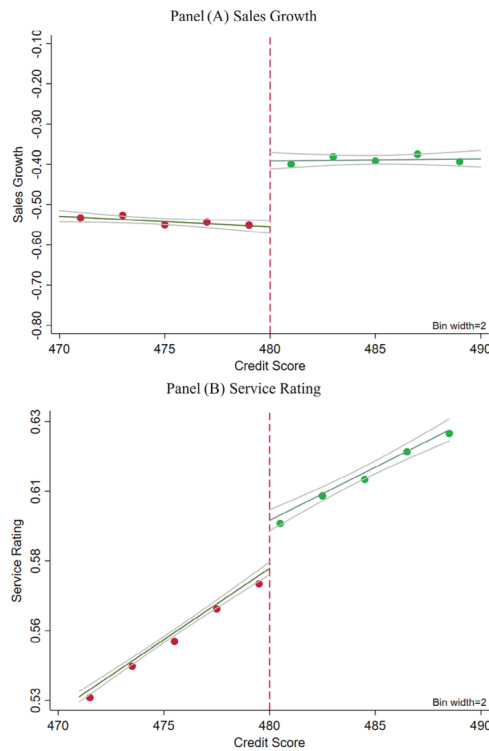


Figure 2. Discontinuity plot on outcome variables. This figure presents discontinuity plots of firm growth from month $t - 1$ to $t + 1$ and CSR ratings at month $t + 1$ against credit scores in month t . The vertical axis is the value of *Sales Growth* and *Service Rating*. The horizontal axis shows credit scores in the local range of [470, 490]. Each dot on the figure represents the average value of the respective outcome measure for firms located in the credit score range with a bandwidth of four. A linear line is fit to the scattered dots on each side of the cutoff score (i.e., 480) and surrounded by a 95% confidence interval in light gray lines. (Color figure can be viewed at wileyonlinelibrary.com)

Read:

- Figure 2 shows a clear jump in sales growth and service rating when firms cross the approval threshold.
- Table III reports the formal 2SLS estimates: sales growth 0.3631, transaction growth 0.2691, product rating 0.0541, service rating 0.0544, and consignment rating 0.0551.
- The first-stage jump in approval is 0.2872, with a Kleibergen-Paap F-statistic of 756.2.

Table III
FinTech Credit Access and Firm Performance

This table shows the FRRD estimates of FinTech credit access on firm performance. We use the 2SLS regression system in equations (1) and (2) to implement the design. In the first stage (Panel A), we regress the credit access dummy, *Credit Approval*, on an indicator variable, *IV* (*Credit Score* ≥ 480), which equals one when the credit score is equal to or greater than 480, and zero otherwise. In the second stage (Panel B), we regress the dependent variable on the instrumented *Credit Approval*. We use the local linear regression model for the credit scores, which range from 460 to 500, and include firm-type and time fixed effects in both stages. The dependent variables are *Sales Growth* and *Transaction Growth* measured from one month before to one month after the credit allocation event, and *Product Rating*, *Service Rating*, and *Consignment Rating*, measured in the month after the credit allocation event. Firm-type and time fixed effects are included. We report *t*-statistics based on standard errors clustered at the firm-type level in parentheses. The Kleibergen-Paap weak instrument statistic is presented in the last row of Panel B. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A. First Stage					
Dependent Var.	<i>Credit Approval</i>				
	(1)				
<i>IV</i> (<i>Credit Score</i> ≥480)	0.2872*** (27.50)				
Polynomials in Credit Score	Yes				
Firm Type FE	Yes				
Time FE	Yes				
<i>N</i>	1,196,887				
Panel B. Second Stage					
Dependent Var.	<i>Sales Growth</i> (1)	<i>Transaction Growth</i> (2)	<i>Product Rating</i> (3)	<i>Service Rating</i> (4)	<i>Consignment Rating</i> (5)
<i>Credit Approval</i> (Instrumented)	0.3631*** (15.97)	0.2691*** (16.31)	0.0541*** (9.17)	0.0544*** (9.62)	0.0551*** (9.90)
Polynomials in Credit Score	Yes	Yes	Yes	Yes	Yes
Firm Type FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
<i>N</i>	1,196,887	1,196,887	1,196,887	1,196,887	1,196,887
Kleibergen-Paap <i>F</i> -stat.	756.2				

Economic message. The magnitudes are the headline result of the paper. These are not tiny local treatment effects. For threshold firms, access to a credit line substantially changes growth and customer experience.

5.4 Table IV: the strongest gains go to younger and more opaque firms

Table IV
FinTech Credit Access and Firm Performance: Information Channel

This table shows the heterogeneous effects of FinTech credit access on firm performance across firms with varying credit information quality. We use smaller firm age (Panel A) and higher dispersion of industry growth (Panel B) as proxies for information asymmetries facing traditional credit and information advantages of FinTech credit. *Age Rank* is the relative firm rank based on age, ranging from zero to one. *High Dispersion* is an indicator variable that equals one if a firm is operating in an industry, in which the standard deviation of sales growth across all firms in the industry prior to the credit allocation event is above the industry median, and zero otherwise. In the first stage, we instrument *Credit Approval* and its interaction with the information advantage proxy as specified in equations (3a) and (3b). In the second stage, we regress performance measures on the instrumented variables in accordance with equation (4). The dependent variables are *Sales Growth*, *Transaction Growth*, *Product Rating*, *Service Rating*, and *Consignment Rating*. We use the local linear regression model for credit scores, which range from 460 to 500, and include firm-type and time fixed effects in both stages. We report *t*-statistics based on standard errors clustered at the firm-type level in parentheses. The Kleibergen-Paap weak instrument statistic is presented in the last row of each panel. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A. Heterogeneous Effect by Firm Age					
Dependent Var.	<i>Sales Growth</i>	<i>Transaction Growth</i>	<i>Product Rating</i>	<i>Service Rating</i>	<i>Consignment Rating</i>
	(1)	(2)	(3)	(4)	(5)
<i>Credit Approval</i> × <i>Age Rank</i>	−0.9153***	−0.5972***	−0.2339***	−0.1806***	−0.1725***
(Instrumented)	(−5.69)	(−4.90)	(−10.98)	(−8.85)	(−8.54)
<i>Credit Approval</i>	0.4733***	0.3413***	0.0867***	0.0810***	0.0807***
(Instrumented)	(22.94)	(23.15)	(10.04)	(10.72)	(10.66)
<i>Age Rank</i>	0.9353***	0.5750***	−0.1630***	−0.2530***	−0.2656***
	(7.15)	(6.02)	(−9.36)	(−14.51)	(−13.35)
Polynomials in Credit Score	Yes	Yes	Yes	Yes	Yes
Firm Type FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
<i>N</i>	1,196,848	1,196,848	1,196,848	1,196,848	1,196,848
Kleibergen-Paap <i>F</i> -stat.	418.4				
Panel B. Heterogeneous Effect by Industry Dispersion in Growth					
Dependent Var.	<i>Sales Growth</i>	<i>Transaction Growth</i>	<i>Product Rating</i>	<i>Service Rating</i>	<i>Consignment Rating</i>
	(1)	(2)	(3)	(4)	(5)
<i>Credit Approval</i> × <i>High Dispersion</i>	0.1910*	0.1297*	0.0456***	0.0381***	0.0353***
(Instrumented)	(1.80)	(1.83)	(5.64)	(4.05)	(3.84)
<i>Credit Approval</i>	0.2160***	0.1685***	0.0192***	0.0253***	0.0281***
(Instrumented)	(2.44)	(2.83)	(5.11)	(5.63)	(6.32)
<i>High Dispersion</i>	−0.2439***	−0.2139***	−0.0319***	−0.0248***	−0.0234***
	(−2.54)	(−3.11)	(−4.88)	(−3.35)	(−3.21)
<i>(Continued)</i>					

(Continued)

Table IV—Continued

Panel B. Heterogeneous Effect by Industry Dispersion in Growth					
Dependent Var.	<i>Sales Growth</i>	<i>Transaction Growth</i>	<i>Product Rating</i>	<i>Service Rating</i>	<i>Consignment Rating</i>
	(1)	(2)	(3)	(4)	(5)
Polynomials in Credit Score	Yes	Yes	Yes	Yes	Yes
Firm Type FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
<i>N</i>	1,196,205	1,196,205	1,196,205	1,196,205	1,196,205
Kleibergen-Paap <i>F</i> -stat.	355.7				

Read:

- In Panel A, the interaction *CreditApproval* × *AgeRank* is negative in all five regressions. Younger firms benefit more.
- In Panel B, the interaction *CreditApproval* × *HighDispersion* is positive for all five outcomes. Firms in harder-to-evaluate industries benefit more.
- The paper calculates that a one-standard-deviation lower age rank yields about 13 percentage points more sales growth and 8 percentage points more transaction growth after approval.

Economic message. This is the main evidence that the effect is not merely “more cash helps everyone.” It is specifically stronger where traditional information problems are most severe.

5.5 Table V: information substitutes for collateral

Table V
FinTech Credit Access and Firm Performance: Information as Substitute for Collateral

This table shows the heterogeneous effects of FinTech credit access on firm performance across vendors with varying degrees of collateral availability. We use asset durability of an industry (Panel A) and estimated real estate property ownership (Panel B) as proxies for collateral availability. *High Durability* is an indicator variable that equals one if a firm operates in a durable goods industry. *Probability Ownership* is an indicator variable that equals one if the estimated probability of the firm owner owning real estate is greater than 0.9. In the first stage, we instrument *Credit Approval* and its interaction with the collateral proxies (i.e., *High Durability* or *Property Ownership*) analogous to equations (3a) and (3b). In the second stage, we regress performance measures on the instrumented variables analogous to equation (4). The dependent variables are *Sales Growth*, *Transaction Growth*, *Product Rating*, *Service Rating*, and *Consignment Rating*. We use the local linear regression model for credit scores, which range from 460 to 500, and include firm-type and time fixed effects in both stages. We report *t*-statistics based on standard errors clustered at the firm-type level in parentheses. The Kleibergen-Paap weak instrument statistic is presented in the last row of each panel. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A. Heterogeneous Effect by Asset Durability					
Dependent Var.	<i>Sales Growth</i>	<i>Transaction Growth</i>	<i>Product Rating</i>	<i>Service Rating</i>	<i>Consignment Rating</i>
	(1)	(2)	(3)	(4)	(5)
<i>Credit Approval</i> $\times$ <i>High Durability</i>	-0.0880***	-0.0736***	-0.0386***	-0.0283**	-0.0308***
(Instrumented)	(-2.17)	(-2.81)	(-3.26)	(-2.31)	(-2.55)
<i>Credit Approval</i>	0.3722***	0.2767***	0.0581***	0.0574***	0.0583***
(Instrumented)	(18.37)	(19.64)	(9.73)	(8.83)	(8.78)
<i>High Durability</i> (absorbed by Firm Type FE)					
Polynomials in Credit Score	Yes	Yes	Yes	Yes	Yes
Firm Type FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
<i>N</i>	1,196,887	1,196,887	1,196,887	1,196,887	1,196,887
Kleibergen-Paap <i>F</i> -stat.	364.7				
Panel B. Heterogeneous Effect by Property Ownership					
Dependent Var.	<i>Sales Growth</i>	<i>Transaction Growth</i>	<i>Product Rating</i>	<i>Service Rating</i>	<i>Consignment Rating</i>
	(1)	(2)	(3)	(4)	(5)
<i>Credit Approval</i> $\times$ <i>Property Ownership</i>	-0.0416**	-0.0248**	-0.0245***	-0.0178***	-0.0167***
(Instrumented)	(-2.14)	(-1.98)	(-4.19)	(-3.04)	(-2.64)
<i>Credit Approval</i>	0.3701***	0.2732***	0.0588***	0.0581***	0.0586***
(Instrumented)	(17.20)	(17.45)	(9.00)	(9.34)	(9.62)
<i>Property Ownership</i>	0.0537***	0.0365***	-0.0059	-0.0153***	-0.0191***
	(3.44)	(3.68)	(-1.29)	(-3.62)	(-3.74)
<i>(Continued)</i>					

(Continued)

Table V—Continued					
Panel B. Heterogeneous Effect by Property Ownership					
Dependent Var.	<i>Sales Growth</i>	<i>Transaction Growth</i>	<i>Product Rating</i>	<i>Service Rating</i>	<i>Consignment Rating</i>
	(1)	(2)	(3)	(4)	(5)
Polynomials in Credit Score	Yes	Yes	Yes	Yes	Yes
Firm Type FE	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes
<i>N</i>	1,196,887	1,196,887	1,196,887	1,196,887	1,196,887
Kleibergen-Paap <i>F</i> -stat.	380.5				

Read:

- Panel A shows that the effect of credit is weaker in durable-goods sectors, where collateral is more naturally available.
- Panel B shows that the effect is weaker for vendors more likely to own real estate.
- In both panels, the interaction terms are negative for growth and ratings, exactly as the collateral-substitution story predicts.

Economic message. Traditional bank credit often leans on collateral when information is weak. The paper's evidence says FinTech can partially replace that collateral channel with data and monitoring.

5.6 Table VI: vendors use credit to advertise, expand product range, and convert more visitors

Read:

- Advertising rises by 0.2679, which corresponds to about a 31% increase.

Table VI
Mechanisms of Vendor Performance Enhancement

This table shows the FRDD estimates of credit access on other firm activities and performance. We use the 2SLS regression system in equations (1) and (2) and report the results for the second stage, where the dependent variables are *Advertisement*, *Product Variety*, *Customer Conversion Rate*, and *Product Collection Rate*. *Adverting* is the natural logarithm of one plus the vendor's monthly expenditures on advertising. *Product Variety* is the natural logarithm of one plus the number of product types for sale. *Customer Conversion Rate* is the ratio of the number of customers that completed transactions to the total number of customers of a firm in a month (scaled by 100). We use the local linear regression model for credit scores, which range from 460 to 500, and include firm-type and time fixed effects in both stages. We report *t*-statistics based on standard errors clustered at the firm-type level in parentheses. The Kleibergen-Paap weak instrument statistic is presented in the last row. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Dependent Var.	<i>Advertising</i> [Ln (1 + Ad Expense)] (1)	<i>Product Variety</i> [Ln (1 + Product Types)] (2)	<i>Customer Conversion Rate</i> (3)
<i>Credit Approval</i> (Instrumented)	0.2679*** (6.68)	0.1233*** (4.99)	0.6979*** (5.56)
Polynomials in Credit Score	Yes	Yes	Yes
Firm Type FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
<i>N</i>	1,196,887	1,196,887	1,196,887
Kleibergen-Paap <i>F</i> -stat.	756.2		

- Product variety rises by 0.1233, about a 13% increase.
- Customer conversion rises by 0.6979, meaning close to one extra buyer per 100 visitors.

Economic message. These outcomes make the growth effect much more believable. The paper is not just showing reduced-form sales jumps; it shows what firms actually do with the new financing capacity.

5.7 Table VII: the effect survives many specification changes

Table VII
Robustness Checks

This table presents robustness results for the baseline specification in Table III using alternative measures, samples, or specifications. All panels except Panel C use specifications analogous to equations (1) and (2). Panel A shows the effect of credit access on sales and service performance, where *Sales Growth* and *Transaction Growth* are measured from the month before to two months after the credit allocation event, and *Product Rating*, *Service Rating*, and *Consignment Rating* are measured two months after the credit allocation event. Panel B uses the *change* in the ratings from the month before the credit allocation event to the month after as the dependent variable. Panel C uses different linear functional forms for credit scores to the right and the left of the cutoff at 480. Panel D adopts a second-order polynomial ($K = 2$) for the credit score controls. Panels E and F use credit scores within the range [465, 495] and [470, 490] as alternative bandwidths, respectively. All other panels use a local range [460, 500] for credit scores. We report *t*-statistics based on standard errors clustered at the industry level in parentheses. The Kleibergen-Paap weak instrument statistic is presented in the last row of each panel. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A. Performance with Three Months' Consecutive Credit Access					
Dependent Var.	<i>Sales Growth</i> (1)	<i>Transaction Growth</i> (2)	<i>Product Rating</i> (3)	<i>Service Rating</i> (4)	<i>Consignment Rating</i> (5)
<i>Credit Approval</i> (Instrumented)	0.8470*** (24.10)	0.5519*** (23.32)	0.0843*** (13.52)	0.0853*** (14.49)	0.0844*** (14.77)
Polynomials in Credit Score	Yes	Yes	Yes	Yes	Yes
Firm Type FE	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes
<i>N</i>	931,599	931,599	881,387	881,387	881,387
Kleibergen-Paap <i>F</i> -stat.	840			896.4	

(Continued)

Table VII—Continued

Panel D. Second-Order Polynomial for the Credit Score Variable					
Dependent Var.	<i>Sales Growth</i> (1)	<i>Transaction Growth</i> (2)	<i>Product Rating</i> (3)	<i>Service Rating</i> (4)	<i>Consignment Rating</i> (5)
<i>Credit Approval</i> (Instrumented)	0.3836*** (16.43)	0.2832*** (16.35)	0.0465*** (7.54)	0.0451*** (8.45)	0.0470*** (8.88)
Polynomials in Credit Score	Yes	Yes	Yes	Yes	Yes
Firm Type FE	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes
<i>N</i>	1,196,887	1,196,887	1,196,887	1,196,887	1,196,887
Kleibergen-Paap <i>F</i> -stat.	1,105.0				

Panel E. Alternative Bandwidth [465, 495]					
Dependent Var.	<i>Sales Growth</i> (1)	<i>Transaction Growth</i> (2)	<i>Product Rating</i> (3)	<i>Service Rating</i> (4)	<i>Consignment Rating</i> (5)
<i>Credit Approval</i> (Instrumented)	0.4159*** (14.43)	0.2957*** (12.91)	0.0513*** (10.08)	0.0506*** (11.68)	0.0515*** (12.13)
Polynomials in Credit Score	Yes	Yes	Yes	Yes	Yes
Firm Type FE	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes
<i>N</i>	862,893	862,893	862,893	862,893	862,893
Kleibergen-Paap <i>F</i> -stat	960.7				

(Continued)

Table VII—Continued

Panel B. Change of Service Ratings					
Dependent Var.	Δ Product Rating (1)	Δ Service Rating (2)	Δ Consignment Rating (3)		
Credit Approval (Instrumented)	0.0323*** (21.36)	0.0325*** (21.40)	0.0349*** (21.36)		
Polynomials in Credit Score	Yes	Yes	Yes		
Firm Type FE	Yes	Yes	Yes		
Time FE	Yes	Yes	Yes		
N	1,014,149	1,014,149	1,014,149		
Kleibergen-Paap F-stat.	602.5				

Panel C. Differential Right and Left Slopes for the Credit Score Variable					
Dependent Var.	Sales Growth (1)	Transaction Growth (2)	Product Rating (3)	Service Rating (4)	Consignment Rating (5)
Credit Approval (Instrumented)	0.3812*** (16.45)	0.2815*** (16.61)	0.0464*** (7.61)	0.0454*** (8.31)	0.0471*** (8.78)
Polynomials in Credit Score	Yes	Yes	Yes	Yes	Yes
Firm Type FE	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes
N	1,196,887	1,196,887	1,196,887	1,196,887	1,196,887
Kleibergen-Paap F-stat.	1,089.0				

(Continued)

Table VII—Continued

Panel F. Alternative Bandwidth [470, 490]					
Dependent Var.	<i>Sales Growth</i> (1)	<i>Transaction Growth</i> (2)	<i>Product Rating</i> (3)	<i>Service Rating</i> (4)	<i>Consignment Rating</i> (5)
<i>Credit Approval</i> (Instrumented)	0.4554*** (10.13)	0.2999*** (8.29)	0.0488*** (9.95)	0.0437*** (9.26)	0.0454*** (9.69)
Polynomials in Credit Score	Yes	Yes	Yes	Yes	Yes
Firm-Type FE	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes
<i>N</i>	553,711	553,711	553,711	553,711	553,711
Kleibergen-Paap <i>F</i> -stat.	849.0				

Read:

Table VIII FinTech Credit Access and Long-term Firm Performance								
This table shows the long-term effects of FinTech credit access on firm performance six months following credit approval. The dependent variable is <i>Sales Growth</i> and <i>Transaction Growth</i> , measured from the month before to six months after a credit allocation event, in columns (1) and (2), respectively, and is <i>Product Rating</i> , <i>Service Rating</i> , <i>Consignment Rating</i> , <i>Advertising</i> , <i>Product Variety</i> , and <i>Customer Conversion Rate</i> , defined in the sixth month following a credit allocation event, in columns (3) to (8), respectively. We use the local linear regression model of the credit scores, which range from 460 to 500, and include firm-type and time fixed effects in both stages. We report <i>t</i> -statistics based on standard errors clustered at the firm-type level in parentheses. The Kleibergen-Paap weak instrument statistic is presented in the last row of each panel. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.								
Dependent Var.	<i>Sales Growth</i>	<i>Transaction Growth</i>	<i>Product Rating</i>	<i>Service Rating</i>	<i>Consignment Rating</i>	<i>Advertising</i> [Ln (1 + Ad Expenses)]	<i>Product Variety</i> [Ln (1 + Product Types)]	<i>Customer Conversion Rate</i>
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Credit Approval</i> (Instrumented)	0.4868*** (8.15)	0.2460*** (5.97)	0.0479*** (7.40)	0.0494*** (7.91)	0.0512*** (8.88)	0.1401*** (4.63)	0.1920*** (6.06)	0.2352*** (9.89)
Polynomials in Credit Score	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm Type FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>N</i>	1,196,887	1,196,887	941,232	941,232	941,232	1,196,887	1,196,887	1,196,887
Kleibergen-Paap <i>F</i> -stat.	756.2		600.3		756.2			

- Panel A shows that requiring three months of continuous credit access yields even larger coefficients: sales growth 0.8470, transaction growth 0.5519, and ratings around 0.084 to 0.085.
- Panel B shows that rating *changes*, not only rating levels, rise after treatment: around 0.032 to 0.035.
- Panels C through F show similar results under differential slopes, second-order polynomials, and narrower bandwidths.
- The placebo random-cutoff test finds virtually no jump in approval at fake cutoffs.

Economic message. Table VII does the heavy credibility work after the baseline. The estimates are not fragile to outcome definitions or functional-form choices, and the identifying jump does not appear at arbitrary cutoffs.

5.8 Table VIII: the effects do not vanish after one month

Read:

- Six months after approval, sales growth is still 0.4868 and transaction growth is still 0.2460.
- Product, service, and consignment ratings remain higher by about 0.048 to 0.051.
- Advertising, product variety, and customer conversion also remain elevated: 0.1401, 0.1920, and 0.2352, respectively.

Economic message. The financing shock has persistence. This is important because it suggests the credit line is not only helping firms smooth one short-term blip; it appears to affect operating decisions over a longer horizon.

6 Short summary

- The paper studies whether access to FinTech credit causally improves entrepreneurial growth for Taobao vendors in China.
- It exploits a fuzzy discontinuity at Ant Group’s internal approval cutoff of 480 and estimates local treatment effects with 2SLS.
- Credit access raises sales, transactions, and customer ratings by economically large amounts.
- The benefits are strongest for younger firms, firms in more opaque industries, and firms with less collateral, which supports an information-advantage interpretation.

- Vendors with credit access spend more on advertising, expand product offerings, and convert more visitors into buyers.
- The results are robust to alternative windows, alternative functional forms, narrower bandwidths, and placebo cutoffs, and many of the effects remain visible six months later.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that access to FinTech credit has a large positive causal effect on the performance of small Chinese e-commerce firms. Near Ant Group's approval threshold, the availability of a credit line materially raises growth, customer satisfaction, and several operating inputs that plausibly produce those gains.

7.2 Economic intuition

The mechanism is not simply "more money helps." The pattern of heterogeneity says more. FinTech credit is especially valuable where traditional banks struggle most: young firms, opaque firms, and firms without strong collateral. Rich transaction data appear to let the lender screen and monitor borrowers in a way that partially substitutes for collateral and relationship lending.

7.3 Takeaway

The paper's broader message is that digital-credit technology can do more than redistribute existing loans. In a credit-friction environment, it can open real growth opportunities for small private firms that are otherwise rationed out of traditional finance.